Max Hirsch

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Research Profile

Final year PhD Applied Math candidate with research at the intersection of numerical analysis, scientific computing, and machine learning. Projects have included inference for multiscale diffusions, convergence and stability for finite element methods solving models of liquid crystals and fluids, and physics-informed machine learning for model order reduction. This includes theoretical analysis of numerical methods, as well as computational projects involving PyTorch and finite element libraries. Developed MLniCS library to integrate PyTorch with RBniCS, a reduced basis library for model order reduction.

Education

Ph.D. in Applied Mathematics, *University of California, Berkeley* *Aug 2023 - Present*
- National Science Foundation Graduate Research Fellowship
- Advisor: Professor Franziska Weber

M.S. in Mathematical Sciences, *Carnegie Mellon University* *Aug 2022 - May 2023*
- Advisor: Professor Franziska Weber

B.S. in Mathematical Sciences, *Carnegie Mellon University* *Aug 2019 – May 2023*

Exchange Semester and Intern, *École Polytechnique Fédérale de Lausanne* *May 2022 - Jan 2023*

Teaching and Mentorship

Research Mentor, University of California, Berkeley *Fall 2023 - Present*
- Cayden Baldwin, Projection Method for Stochastic Navier-Stokes Equations (Summer 2026)
- Leon Mikulinsky, Q-Tensor Bayesian Inverse Problem (Summer 2024 - Spring 2025)
- Yijin (Cici) Wang, Physics-Informed Neural Networks for Q-Tensors (Summer 2024)
- Jose Luis Alarcon, Discovering Numerical Schemes from Data (Fall 2023)

Graduate Student Instructor, University of California, Berkeley *Spring 2025 - Present*
- Calculus I, Fall 2026
- Abstract Linear Algebra, Spring 2025

Teaching Assistant, Carnegie Mellon University *Fall 2020 - Spring 2023*
- Vector Calculus for Computer Scientists, Spring 2023
- Matrices and Linear Transformations, Fall 2021
- Integration and Approximation, Spring 2021
- Matrices and Linear Transformations, Fall 2020

Publications

Preprints

Convergence and Sketching-Based Efficient Computation of Neural Tangent Kernel Weights in Physics-Based Loss *November 2025*
Max Hirsch, Federico Pichi
Submitted to Journal of Computational Physics arXiv preprint arXiv:2511.15530

Nonparametric estimation of homogenized invariant measures from multiscale data via Hermite expansion *October 2025*
Jaroslav Borodavka, Max Hirsch, Sebastian Krumscheid, Andrea Zanoni
Submitted to Journal of Nonlinear Science arXiv preprint arXiv:2510.25521

Journal Articles

A convergent finite element scheme for the Q-tensor model of liquid crystals subjected to an electric field *March 2026*
Max Hirsch, Franziska Weber
Journal of Numerical Mathematics 34 (3), 487–528

Neural empirical interpolation method for nonlinear model reduction *November 2025*
Max Hirsch, Federico Pichi, Jan Hesthaven
SIAM Journal on Scientific Computing 47 (6), C1264-C1293

The zero inertia limit for the Q-tensor model of liquid crystals: analysis and numerics *September 2025*
Max Hirsch, Franziska Weber, Yukun Yue
ESAIM: Mathematical Modelling and Numerical Analysis 59 (5), 2515-2556

Stochastic gradient descent in continuous time for drift identification in multiscale diffusions *November 2024*
Max Hirsch, Andrea Zanoni
ESAIM: Mathematical Modelling and Numerical Analysis 58 (6), 2387-2414

Ongoing Projects

Convergence of a fully discrete discontinuous Galerkin scheme for the stochastic 2D Euler equations
Max Hirsch, Franziska Weber

Nonlinear generalized Kolmogorov widths for model order reduction
Max Hirsch, Federico Pichi

Kernel empirical interpolation method for model order reduction with error estimates
Max Hirsch

Talks

- Equa∂iff Conference 2026, *Convergence and Sketching-Based Efficient Computation of Neural Tangent Kernel Weights in Physics-Based Loss* (June 2026)
- Applied PDE Seminar at UC Berkeley, *Convergence of a Discontinuous Galerkin Scheme for the 2D Stochastic Incompressible Euler Equations* (April 2026)
- Fluid Dynamics Lunch and Learn Seminar at UC Berkeley, *Computing Bifurcation Diagrams for Fluids* (November 2025)
- European Conference on Numerical Mathematics and Advanced Applications (ENUMATH), *Learning the Drift in Multiscale Diffusions* (September 2025)
- mathLab at Scuola Internazionale Superiore di Studi Avanzati (SISSA), *Neural empirical interpolation method for nonlinear model reduction* (July 2025)
- High-Dimensional Scientific Computing Seminar at UC Berkeley, *Neural empirical interpolation method for nonlinear model reduction* (March 2025)
- Computational Math Seminar at UC Berkeley, *Bayesian Inverse Problem for the Q-Tensor Model of Liquid Crystals* (October 2024)
- Applied PDE Seminar at UC Berkeley, *Neural empirical interpolation method for nonlinear model reduction* (August 2024)
- European Congress on Computational Methods in Applied Sciences and Engineering (ECCOMAS), *Neural empirical interpolation method for nonlinear model reduction* (June 2024)

- EPFL Chair of Computational Mathematics and Simulation Sciences Seminar, *Applications of MLniCS : A Library for Physics-Informed Machine Learning for Parametric Problems* (November 2022)
- Finite Element Circus at The Pennsylvania State University, *Finite Elements for the Q-Tensor Flow of Liquid Crystals* (November 2021)

Work Experience

Research Assistant *Remote*
Professor Jay Newby, University of Alberta Department of Mathematical and Statistical Sciences *May 2020 – Aug 2021*

- Estimated particle densities in videos using convolutional neural networks, achieving 6% error in estimates of the total number of particles from these densities.
- Used Monte Carlo methods for maximum likelihood estimation of the diffusion constant arising from the master equation describing particle motion.

Data Science Intern *Hickory, NC*
Matrix Retail *May 2019 – Aug 2019*

- Created a new scalable sales forecasting algorithm for high-dimensional data which automatically tunes parameters, trains, and forecasts with bootstrapped prediction intervals for over 1000 retail stores to reduce the time of the former forecasting algorithm by 300% without sacrificing accuracy.
- Enabled more efficient evaluation of forecasts by creating a graphical user interface for running the forecasting algorithm and visualizing the resulting forecasts and prediction intervals.

Projects

MLniCS Code Library *May 2022 - January 2023*

- Developed MLniCS, a code library for building physics-informed neural networks for reduced order parameterized partial differential equations.
- MLniCS Repository

Tetris Reinforcement Learning (Project Leader) *Spring 2020 - Fall 2020*

- Led a Data Science Club project group in which we trained a reinforcement learning model with the REINFORCE algorithm and Monte Carlo tree search to play Tetris.

Honors and Awards

- March 2026 - Grant for SLMath Summer Graduate School: Mathematics for Machine Learning
- May 2024 - Conference Travel Grant, UC Berkeley Graduate Division
- April 2023 - Mathematics Prize, CMU Dept. of Mathematical Sciences
- April 2023 - Nominated to Phi Beta Kappa Society
- March 2023 - National Science Foundation Graduate Research Fellowship
- May 2022 - Richard A. Moore Award, CMU Dept. of Mathematical Sciences
- April 2022 - Grant to attend HYP2022 Conference on Hyperbolic Problems
- March 2022 - ThinkSwiss Research Scholarship
- February 2022 - EPFL Research Internship Scholarship of Excellence
- May 2021 - Richard A. Moore Award, CMU Dept. of Mathematical Sciences
- November 2019 - CMU Data Science Club November Challenge Winner
- May 2019 - CMU Quantitative Social Science Scholar (QSSS)
- March 2019 - MathWorks Math Modeling Challenge (M3C) Honorable Mention, 1st in Technical Computing
- February 2019 - Mathematical Contest in Modeling (MCM) Honorable Mention

Skills

- **Programming:** Python (PyTorch, NumPy, TensorFlow, scikit-learn), C, MATLAB, R, Git, LaTeX.
- **Languages:** English (native), French (proficient).